

Md. Nasfiqul Islam Antu

Undergraduate Student, Civil & Environmental Engineering, IUT

Phone: (+880) 1773-890624 | **Email:** nasfiqul.antu@gmail.com

Address: Mirpur 14, Dhaka, Bangladesh

LinkedIn: linkedin.com/in/nasfiqul-antu | **GitHub:** github.com/AntuHere

Google Scholar Profile: scholar.google.com/citations?user=Wtr485cAAAAJ

ABOUT ME

I am a final-year Civil and Environmental Engineering undergraduate at the Islamic University of Technology (IUT), with a strong interest in transportation and traffic engineering. I have co-authored seven peer-reviewed journal articles on climate change, food systems and the environment, where I contributed to field data collection, data analysis and literature review. I am currently working on three independent research projects that apply computer vision (YOLO, OpenCV, Roboflow) to traffic video from Dhaka and statistical analysis to survey data. I am seeking a research assistant role to strengthen my skills in data-driven transportation research, machine learning and academic writing, and to contribute to peer-reviewed publications.

EDUCATION

B.Sc. in Civil and Environmental Engineering — Expected: October 2027
Islamic University of Technology (IUT), Gazipur, Bangladesh — Currently in 4th Year (7th Semester)

Relevant Coursework: Transportation and Traffic Engineering; Civil Engineering Data Analysis; Pavement and Railway Engineering

Higher Secondary Certificate (HSC), Science — 2022
Dhaka College, Dhaka — GPA: 5.00 / 5.00

Secondary School Certificate (SSC), Science — 2020
BIAM Model School and College, Bogura — GPA: 5.00 / 5.00

RESEARCH EXPERIENCE

Publications (Peer-Reviewed Journal Articles)

My contribution: field data collection, data analysis and, in several papers, literature review.

1. Islam, M. A., Islam, M. S., Sufian, M. A., & **Islam, M. N.** (2026). Impacts of climate change on agricultural transformation in Satkhira District, Bangladesh: Challenges of livelihood diversification. Journal of Agriculture and Ecology Research International, 27(4), 76–89. https://doi.org/10.9734/jaeri/2026/v27i4766
2. Islam, M. A., & **Islam, M. N.** (2026). A study on food loss and waste (FLW) and its environmental impact at the rural market in Faridpur District, Bangladesh. European Journal of Nutrition & Food Safety, 18(5), 57–71. https://doi.org/10.9734/ejnfs/2026/v18i52036
3. Islam, M. A., & **Islam, M. N.** (2026). Smallholder farmers' participation in the agricultural market systems in Rangpur District, Bangladesh: Challenges and opportunities. Asian Journal of Agricultural and Horticultural Research, 13(2), 88–102. https://doi.org/10.9734/ajahr/2026/v13i2458
4. Islam, M. A., & **Islam, M. N.** (2026). Perceptions of vegetable and fruit retailers on food loss and waste (FLW) and its environmental impacts in rural markets of Faridpur, Bangladesh. International Journal of Environment and Climate Change, 16(4), 174–185. https://doi.org/10.9734/ijecc/2026/v16i45353
5. Islam, M. A., & **Islam, M. N.** (2026). Smallholder farmers' perceptions of climate-smart practices in oilseed and pulse production: Evidence from Barguna district, Bangladesh. Archives of Agriculture and Environmental Science, 11(2), 168–175. https://dx.doi.org/10.26832/24566632.2026.110203
6. Islam, M. A., Islam, M. S., & **Islam, M. N.** (2025). Economic empowerment and livelihood transformation of indigenous women in Bandarban District, Bangladesh. South Asian Journal of Social Studies and Economics, 22(12), 242–255. https://doi.org/10.9734/sajsse/2025/v22i121234
7. Islam, M. A., & **Islam, M. N.** (2025). Private-sector agro-input companies contribute to nutrition-sensitive agriculture in southern Bangladesh: Evidence from Faridpur and Khulna. European Journal of Nutrition and Food Safety, 17(10), 310–326. https://doi.org/10.9734/ejnfs/2025/v17i101886 | Also on HAL: https://hal.science/hal-05602350

Ongoing Research

Rickshaw Passenger Pickup/Drop-off and Stopping-Duration Analysis Ongoing

Independent research, in collaboration with a faculty member · Dhaka (Mirpur, ECB Chattar, Dhanmondi)

- Prepared and annotated roadside video data from three Dhaka locations in Roboflow, and trained the detection dataset.
- Developing and fine-tuning a YOLO-based model to detect rickshaws and capture passenger pickup/drop-off and stopping events.
- Carried out the literature review on non-motorised vehicle stopping behaviour and curbside activity.
- Status: data collection complete; model fine-tuning and analysis in progress.

Lateral Movement of Vehicles in Mixed Traffic Ongoing

Independent collaborative research · Bangla Motor Intersection, Dhaka

- Developing a YOLO-based computer-vision workflow to detect vehicles and study their lateral movement in non-lane-based, heterogeneous traffic.
- Conducting the literature review on lateral behaviour and lane discipline in mixed traffic streams.
- Status: model development ongoing; video data collection at Bangla Motor scheduled.

Road Construction and Demolition Waste (RCDW) on Soil Pollution and Environmental Impacts Ongoing

Independent collaborative research · Sirajganj, Bangladesh

- Questionnaire-based social survey on the soil-pollution and environmental impacts of road construction and demolition waste.
- Responsible for the literature review and the statistical analysis of the survey data.
- Status: data collection complete; analysis in progress.

Research Interests

Transportation & Traffic Engineering · Intelligent Transportation Systems (ITS) · Computer Vision for Traffic Data Extraction · Mixed / Heterogeneous Traffic Behaviour · Road Safety · Machine Learning & Predictive Modelling · Environmental Sustainability & Climate Change

SKILLS AND EXPERTISE

Programming & Data Analysis	Python (Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook), R, Microsoft Excel
Statistical Analysis	Descriptive statistics; hypothesis testing (t-test, ANOVA, chi-square, correlation analysis); linear, multiple and logistic regression
Machine Learning	Familiarity with scikit-learn, regression and Random Forest. Basic familiarity with Decision Tree, XGBoost, LightGBM, SVM, KNN and Naive Bayes.
Explainable AI (XAI)	Basic familiarity with SHAP, LIME and feature-importance analysis
Computer Vision & Deep Learning	YOLO object detection (training and fine-tuning), OpenCV, Roboflow (annotation and dataset management), PyTorch / TensorFlow
Research Methods	Field and questionnaire survey data collection, quantitative data analysis, literature review, academic writing
GIS & Traffic Simulation	Basic GIS (short course, Institute of Water Modelling); currently learning SUMO and PTV VISSIM
Tools & Documentation	Microsoft Word, PowerPoint, LaTeX, Mendeley / Zotero, Git and GitHub

LANGUAGE PROFICIENCY

Bangla: Native
English: Proficient (reading, writing and speaking)